MARS MISSION DESIGN LAB / STUDENT PORTFOLIO

★ Engineering level 4

Student / team: Student Engineering Team

Mission challenge

Level 4 · Flight Director
Collect and cache two samples, send a relay signal, then launch their Earth-return capsule. Include at least 5 obstacles to avoid.

Engineering rank

Flight Director · 4/4 levels passed in this browser
This design passed its Python mission checks.

Mission design screenshot



Design captured for this proposal. Dashed line: planned route. Solid line: latest coded traverse.

Mission resources and testing

> Build credits: 910 / 1,000 | Terrain obstacles: 5 / 5 minimum
> Planned route: 100 map units | Planned energy: 7 / 100
> Latest test: Mission complete
> Actual code run: 500 map units | 33 / 100 energy | 11 commands

Our science question

Not yet written.

Our engineering decisions

Our completed mission engineering explanation.

Our Python rover program

Student-written program, including comments and indentation.

```
roll(100, 90, 0.7812500)
collect()
roll(100, 90, 0.7812500)
collect()
roll(100, 90, 0.7812500)
collect()
roll(100, 90, 0.7812500)
transmit()
roll(100, 90, 0.7812500)
launch()
stop()
```

Python demonstration checks

PASS: Collect two different rock samples
PASS: Deposit two samples at a cache
PASS: Send orange-blue at a relay
PASS: Launch the prepared samples on a simulated Earth return
PASS: Include at least 5 terrain obstacles (5/5 placed)
PASS: Drive safely through the terrain
PASS: Meet build and actual code energy limits
PASS: Demonstrate a successful Python program

Placed equipment and obstacles

Object	Type	Position (X, Y)	Credits
Mars rover	Mars rover	100, 500	300
Landing zone	Landing zone	200, 500	80
Comms relay	Comms relay	500, 500	130
Rock field	Rock field	100, 100	0
Rock field	Rock field	260, 100	0
Rock field	Rock field	420, 100	0
Rock field	Rock field	580, 100	0
Rock field	Rock field	740, 100	0
Rock sample	Rock sample	200, 500	60
Rock sample	Rock sample	300, 500	60
Sample cache	Sample cache	400, 500	100
Earth-return capsule	Earth-return capsule	600, 500	180

Planned flight path

1. 200, 500

Independent classroom simulation. Imagined Mars terrain and simplified build credits, energy and mission actions.